

EE114 - Homework 2

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Problem 1.5.1

Given the model of handoffs and call lengths in Problem 1.4.1,

- (a) What is the probability that a brief call will have no handoffs?

$$P[H_0|B] = \frac{P[H_0 \cap B]}{P[B]} = \frac{0.4}{0.4+0.1+0.1} = \frac{0.4}{0.6} = \frac{2}{3} \approx \mathbf{0.667}$$

- (b) What is the probability that a call with one handoff will be long?

$$P[L|H_1] = \frac{P[L \cap H_1]}{P[H_1]} = \frac{0.1}{0.1+0.1} = \frac{0.1}{0.2} = \mathbf{0.5}$$

- (c) What is the probability that a long call will have one or more handoffs?

$$P[(H_1 \cup H_2)|L] = \frac{P[(H_1 \cup H_2) \cap L]}{P[L]} = \frac{P[H_1 \cap L] + P[H_2 \cap L]}{P[L]} = \frac{0.1+0.2}{0.1+0.1+0.2} = \frac{0.3}{0.4} = \mathbf{0.75}$$

Problem 1.5.3

You have a shuffled deck of three cards: 2, 3, and 4. You draw one card. Let C_i denote the event that card i is picked. Let E denote the event that card chosen is an even-numbered card.

- (a) What is $P[C_2|E]$, the probability that the 2 is picked given that an even-numbered card is chosen?

The event E (even card) is $\{2, 4\}$, so $P[E] = 2/3$. The intersection $C_2 \cap E$ is simply $\{2\}$, so $P[C_2 \cap E] = 1/3$.

$$P[C_2|E] = \frac{P[C_2 \cap E]}{P[E]} = \frac{1/3}{2/3} = \frac{1}{2}$$

- (b) What is the conditional probability that an even-numbered card is picked given that the 2 is picked?

Given C_2 (card is 2), the card is definitely even, so $E \cap C_2 = \{2\}$ and $P[E \cap C_2] = 1/3$. Also $P[C_2] = 1/3$.

$$P[E|C_2] = \frac{P[E \cap C_2]}{P[C_2]} = \frac{1/3}{1/3} = 1$$

Problem 1.5.4

From Problem 1.4.3, what is the conditional probability of yy , that a pea plant has two dominant genes given the event Y that it has yellow seeds?

From Problem 1.4.3, we have the following genotype probabilities:

- $P[YY] = 1/4$ (Two dominant genes)
- $P[Yy] = 1/2$ (Hybrid)
- $P[yy] = 1/4$ (Two recessive genes)

The event Y (yellow seeds) occurs for genotypes $\{YY, Yy\}$. The probability of Y is:

$$P[Y] = P[YY] + P[Yy] = 1/4 + 1/2 = 3/4$$

We want to find the probability of having two dominant genes (YY) given Y . The intersection $\{YY\} \cap Y$ is simply $\{YY\}$, so $P[YY \cap Y] = 1/4$.

$$P[YY|Y] = \frac{P[YY \cap Y]}{P[Y]} = \frac{1/4}{3/4} = \frac{1}{3}$$

Problem 1.6.1

Is it possible for A and B to be independent events yet satisfy $A = B$?

Yes, but only in trivial cases.

The definition of independence for events A and B is:

$$P[A \cap B] = P[A]P[B]$$

If $A = B$, then $A \cap B = A \cap A = A$. The condition becomes:

$$P[A] = P[A] \cdot P[A] = (P[A])^2$$

Let $x = P[A]$. Then $x = x^2 \implies x^2 - x = 0 \implies x(x - 1) = 0$.

The solutions are $x = 0$ and $x = 1$.

Thus, A and B can be independent and equal if and only if $P[A] = 0$ or $P[A] = 1$.

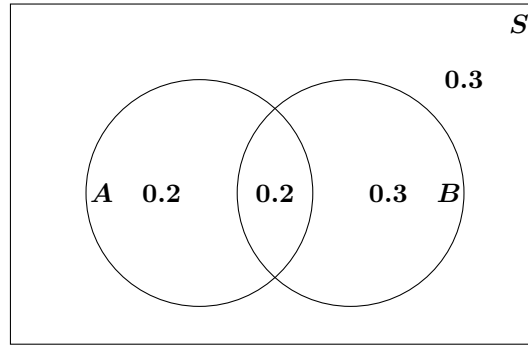
Problem 1.6.2

Use a Venn diagram in which the event areas are proportional to their probabilities to illustrate two events A and B that are independent.

We select probabilities $P[A] = 0.4$ and $P[B] = 0.5$. For independence, $P[A \cap B] = P[A]P[B] = 0.4 \times 0.5 = 0.2$.

This implies $P[A \cap B^c] = 0.4 - 0.2 = 0.2$ and $P[B \cap A^c] = 0.5 - 0.2 = 0.3$.

The area outside both events is $1 - (0.2 + 0.2 + 0.3) = 0.3$.



Since the intersection area (0.2) equals the product of the individual probabilities (0.4×0.5), the events A and B are visually independent.

Problem 1.6.3

In an experiment, A, B, C , and D are events with probabilities $P[A] = 1/4$, $P[B] = 1/8$, $P[C] = 5/8$, and $P[D] = 3/8$. Furthermore, A and B are disjoint, while C and D are independent.

- (a) Find $P[A \cap B]$, $P[A \cup B]$, $P[A \cap B^c]$, and $P[A \cup B^c]$.

Given A, B are disjoint:

- $P[A \cap B] = 0$.
- $P[A \cup B] = P[A] + P[B] = 1/4 + 1/8 = 3/8$.
- $P[A \cap B^c] = P[A] = 1/4$ (Since A and B disjoint implies $A \subseteq B^c$).
- $P[A \cup B^c] = P(A) + P(B^c) - P(A \cap B^c) = 1/4 + (1 - 1/8) - 1/4 = 7/8$.

- (b) Are A and B independent?

No. Check: $P[A \cap B] = 0$. $P[A]P[B] = (1/4)(1/8) = 1/32$.

Since $0 \neq 1/32$, they are not independent.

- (c) Find $P[C \cap D]$, $P[C \cap D^c]$, and $P[C^c \cap D^c]$.

Given C, D are independent:

- $P[C \cap D] = P[C]P[D] = (5/8)(3/8) = 15/64$.
- $P[C \cap D^c] = P[C]P[D^c] = (5/8)(1 - 3/8) = (5/8)(5/8) = 25/64$.
- $P[C^c \cap D^c] = P[C^c]P[D^c] = (1 - 5/8)(1 - 3/8) = (3/8)(5/8) = 15/64$.

- (d) Are C^c and D^c independent?

Yes. If C and D are independent, their complements are also independent.

Verification: $P[C^c \cap D^c] = 15/64$. $P[C^c]P[D^c] = (3/8)(5/8) = 15/64$.

Problem 1.6.4

In an experiment, A, B, C , and D are events with probabilities $P[A \cup B] = 5/8$, $P[A] = 3/8$, $P[C \cap D] = 1/3$, and $P[C] = 1/2$. Furthermore, A and B are disjoint, while C and D are independent.

- (a) Find $P[A \cap B]$, $P[B]$, $P[A \cap B^c]$, and $P[A \cup B^c]$.

Given A, B are disjoint:

- $P[A \cap B] = 0$.
- $P[A \cup B] = P[A] + P[B] \implies 5/8 = 3/8 + P[B] \implies P[B] = 2/8 = 1/4$.
- $P[A \cap B^c] = P[A] = 3/8$ (Since $A \subseteq B^c$).
- $P[A \cup B^c] = P[A] + P[B^c] - P[A \cap B^c] = 3/8 + (1 - 1/4) - 3/8 = 3/4$.

- (b) Are A and B independent?

No. $P[A \cap B] = 0$, but $P[A]P[B] = (3/8)(1/4) = 3/32$. Since $0 \neq 3/32$, they are not independent.

- (c) Find $P[D]$, $P[C \cap D^c]$, $P[C^c \cap D^c]$, and $P[C|D]$.

Given C, D are independent:

- $P[C \cap D] = P[C]P[D] \implies 1/3 = (1/2)P[D] \implies P[D] = 2/3$.
- $P[C \cap D^c] = P[C]P[D^c] = (1/2)(1 - 2/3) = 1/6$.
- $P[C^c \cap D^c] = P[C^c]P[D^c] = (1 - 1/2)(1/3) = 1/6$.
- $P[C|D] = P(C) = 1/2$ (due to independence).

- (d) Find $P[C \cup D]$ and $P[C \cup D^c]$.

- $P[C \cup D] = P[C] + P[D] - P[C \cap D] = 1/2 + 2/3 - 1/3 = 5/6$.
- $P[C \cup D^c] = P[C] + P[D^c] - P[C \cap D^c] = 1/2 + 1/3 - 1/6 = 4/6 = 2/3$.

- (e) Are C and D^c independent?

Yes. Since C and D are independent, C and D^c are also independent.

Problem 1.6.5

In an experiment with equiprobable outcomes, the event space is $S = \{1, 2, 3, 4\}$ and $P[s] = 1/4$ for all $s \in S$. Find three events in S that are pairwise independent but are not independent. (Note: Pairwise independent events meet the first three conditions of Definition 1.8).

Consider the events $A = \{1, 2\}$, $B = \{1, 3\}$, and $C = \{1, 4\}$.

- $P[A] = P[B] = P[C] = 2/4 = 1/2$.
- $P[A \cap B] = P[\{1\}] = 1/4 = P[A]P[B]$. (Pairwise Independent)
- $P[A \cap C] = P[\{1\}] = 1/4 = P[A]P[C]$. (Pairwise Independent)
- $P[B \cap C] = P[\{1\}] = 1/4 = P[B]P[C]$. (Pairwise Independent)

Check mutual independence:

$$P[A \cap B \cap C] = P[\{1\}] = 1/4$$

$$P[A]P[B]P[C] = (1/2)^3 = 1/8$$

Since $1/4 \neq 1/8$, the events are pairwise independent but not mutually independent.

Problem 1.6.7

For independent events A and B , prove that

- (a) A and B^c are independent.

We need to show $P[A \cap B^c] = P[A]P[B^c]$.

$$\begin{aligned} P[A] &= P[A \cap B] + P[A \cap B^c] \quad (\text{Total Probability}) \\ P[A \cap B^c] &= P[A] - P[A \cap B] \\ &= P[A] - P[A]P[B] \quad (\text{Since } A, B \text{ independent}) \\ &= P[A](1 - P[B]) \\ &= P[A]P[B^c]. \quad (\text{Proven}) \end{aligned}$$

- (b) A^c and B are independent.

We need to show $P[A^c \cap B] = P[A^c]P[B]$.

$$\begin{aligned} P[B] &= P[A \cap B] + P[A^c \cap B] \\ P[A^c \cap B] &= P[B] - P[A \cap B] \\ &= P[B] - P[A]P[B] \quad (\text{Since } A, B \text{ independent}) \\ &= P[B](1 - P[A]) \\ &= P[B]P[A^c]. \quad (\text{Proven}) \end{aligned}$$

- (c) A^c and B^c are independent.

We need to show $P[A^c \cap B^c] = P[A^c]P[B^c]$.

$$\begin{aligned} P[A^c \cap B^c] &= P[(A \cup B)^c] \quad (\text{De Morgan's Law}) \\ &= 1 - P[A \cup B] \\ &= 1 - (P[A] + P[B] - P[A \cap B]) \\ &= 1 - P[A] - P[B] + P[A]P[B] \quad (\text{Independence}) \\ &= (1 - P[A]) - P[B](1 - P[A]) \\ &= (1 - P[A])(1 - P[B]) \\ &= P[A^c]P[B^c]. \quad (\text{Proven}) \end{aligned}$$